GyeongNam Kang, M.S.
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Research Portfolio — Click to view research and publications

1. RESEARCH INTERESTS

- Research on generative data augmentation for construction hazard detection began as an undergraduate project and later developed into a journal article (4.2 [01]).
- Limitations of object detection in assessing safety compliance led to the exploration of vision–language models for interpreting site conditions against safety rules (4.3 [06]).
- Generative AI applications were further extended beyond training data to optical illusion-based safety signs combining visual effects with safety messages (4.2 [03]).
- Variability across generation tools, settings, and augmentation volumes led to the development of an agentic system linking generation, validation, training, and evaluation (4.2 [04]).
- These studies strengthened my ability to apply and integrate diverse AI models and techniques for generating and interpreting visual information in construction.
- Building on this experience, I am interested in treating visual information as one modality and combining it with language, sensor data, and contextual information to better understand construction processes and support management decisions.

2. EDUCATION

M.S. Architectural Engineering (Construction Management and Engineering) ..... 03/2025 – 08/2026
- Pusan National University, Busan, South Korea (Advisor: JungHo Jeon)
- Thesis: LLM-Based Multi-Agent Architecture for Improving Construction Hazard Detection
- GPA: 4.50/4.50

B.S. Architectural Engineering ..... 03/2019 – 02/2025
- Pusan National University, Busan, South Korea (Advisor: JungHo Jeon)
- GPA: 4.14/4.50

3. INDUSTRY EXPERIENCE

Full-time Researcher ..... 09/2026 – Present
- Research Institute of Industrial Technology, Pusan National University, Busan, South Korea
- Main Tasks: AI-based construction safety research, research proposal development, and preparation of journal manuscripts and industrial safety camera guidelines

Construction Engineering Intern ..... 06/2024 – 08/2024
- Hyundai Engineering & Construction Co., Ltd., South Korea
- Main Tasks: Crack and waterproofing inspections, mortar mixing and flow tests, and field coordination
- Main Project: Hillstate Hwangseong Apartment Construction, Gyeongju

4. RESEARCH EXPERIENCE

4.1 Research Projects (4 funded projects)

Effectiveness Validation and Guideline Development for
Fixed AI-Based Human Detection Cameras ..... 06/2026 – 11/2026
- Supported by Occupational Safety and Health Research Institute, Korea Occupational Safety and Health Agency (KOSHA)
- Role: Participating Researcher, Pusan National University

Construction Safety Management Agents Based on Human Foundation Models ..... 03/2026 – Present
- Supported by National Research Foundation of Korea (NRF)
- Role: Graduate Research Assistant, Pusan National University

Safety Management Platform Development through
Generative AI-Based Data Augmentation ..... 03/2025 – Present
- Supported by National Research Foundation of Korea (NRF)
- Role: Graduate Research Assistant, Pusan National University

Development of the BIM-based Decision Support System for
Green Building Planning with Each Design Phase ..... 06/2024 – 02/2026
- Supported by National Research Foundation of Korea (NRF)
- Role: Research Assistant, Pusan National University

4.2 Peer-Reviewed Journal Articles (3 published and 1 planned submission)

[04] **Kang, G.**, & Jeon, J. (2026). "PAAS: Partial Autonomous Agentic System for Maximizing Computer Vision-based Construction Hazard Detection". Computer-Aided Civil and Infrastructure Engineering. *(Submission planned for November 2026)*

[03] **Kang, G.**, Lee, J., Kim, N., & Jeon, J. (2026). "A generative AI framework for developing optical illusion-based construction safety signs". Developments in the Built Environment, 27, 100978. https://doi.org/10.1016/j.dibe.2026.100978

[02] **Kang, G.**, Cho, S., Kang, M., Kim, H., Min, J., & Jeon, J. (2026). "Enhancing Drowsiness Detection in Construction Workers Through Generative Image Data and Computer Vision". Journal of the Architectural Institute of Korea, 42(3), 349-356. https://doi.org/10.5659/JAIK.2026.42.3.349

[01] Lee, Y., **Kang, G.**, Kim, J., Yoon, S., & Jeon, J. (2025). "Generative AI-driven data augmentation for enhanced construction hazard detection". Automation in Construction, 177, 106317. https://doi.org/10.1016/j.autcon.2025.106317

4.3 Conference Papers (6 published and 2 forthcoming)

[08] **Kang, G.**, Kim, K., Cho, S., Kim, H., & Jeon, J. (2026). "Improving Construction Site Hazardous Object Detection Performance through Partial Autonomous Agentic AI-based Data Augmentation". In Proceedings of the 2026 Annual Conference of the Korea Institute of Construction Engineering and Management. *(Presentation and publication planned for November 2026)*

[07] **Kang, G.**, Song, K., Park, H., & Jeon, J. (2026). "LLM-Orchestrated Multi-Agent System for Data Augmentation in Construction Hazard Detection". In Proceedings of the ASCE International Conference on Computing in Civil Engineering (i3CE 2026). Reston, VA: ASCE. *(forthcoming)*

[06] **Kang, G.**, Lee, J., Lee, J., Kim, H., Park, H., & Jeon, J. (2025). "Visual Reasoning for Construction Safety Monitoring". In International Conference on Construction Engineering and Project Management (pp. 438-445). Korea Institute of Construction Engineering and Management. **(2025 ICCEPM Best Paper Award).** https://doi.org/10.6106/ICCEPM.2025.0438

[05] Jeong, K., Sung, J., **Kang, G.**, Park, K., Park, H., & Jeon, J. (2025). "Verification of Construction Site Safety Compliance Using a VQA Model: Performance Analysis across Situational Contexts". In Proceedings of the 2025 Annual Conference of the Korea Institute of Construction Engineering and Management (pp. 45-46). **(2025 KICEM Annual Conference Best Paper Award).**

[04] Sung, J., Jeong, K., Lee, Y., **Kang, G.**, Park, H., & Jeon, J. (2025). "Multi-Class Classification of Construction Workers' Stress States Using rPPG". In Proceedings of the 2025 Fall Conference of the Korea Institute of Building Construction, 25(2) (pp. 71-72). **(2025 Korea Institute of Building Construction Fall Conference Outstanding Academic Paper Award).**

[03] Lee, Y., **Kang, G.**, Cho, W., Sung, J., Park, H., & Jeon, J. (2025). "Development of an AI-Based Generative Image Quality Assessment Framework for Construction Safety Management". In Proceedings of the Architectural Institute of Korea, 45(1) (pp. 669-670).

[02] Lee, Y., **Kang, G.**, Cho, W., Cho, S., & Jeon, J. (2024). "A Generative AI-Based Data Augmentation Framework for Improving Safety Helmet Detection on Construction Sites". In Proceedings of the Busan-Ulsan-Gyeongnam Chapter of the Architectural Institute of Korea (pp. 357-358).

[01] **Kang, G.**, Lee, Y., Cho, W., Cho, S., Yoon, S., & Jeon, J. (2024). "Generative AI-Based Data Augmentation for Improving Computer Vision Hazard Identification". In Proceedings of the Architectural Institute of Korea, 44(2) (pp. 1184-1187). **(2024 Architectural Institute of Korea Fall Conference Outstanding Presentation Paper Award).**

4.4 Research Service

- Reviewer of Engineering Applications of Artificial Intelligence (EAAI), ELSEVIER (2026 – Present)

5. TEACHING EXPERIENCE

5.1 Courses

Teaching Assistant, Pusan National University ..... Spring 2026
- Course Name: Advanced Construction IT
- Instructor: JungHo Jeon

Teaching Assistant, Pusan National University ..... Fall 2025
- Course Name: Smart Construction Safety Technology
- Instructor: JungHo Jeon

Teaching Assistant, Pusan National University ..... Fall 2025
- Course Name: Cost Planning and Management
- Instructor: Youngki Huh

Teaching Assistant, Pusan National University ..... Spring 2025
- Course Name: Advanced Construction Safety Management
- Instructor: JungHo Jeon

Teaching Assistant, Pusan National University ..... Spring 2025
- Course Name: Construction Project Management
- Instructor: Youngki Huh

6. HONORS AND AWARDS

2025 ICCEPM Best Paper Award 11/2025
2025 KICEM Annual Conference Best Paper Award 11/2025
Outstanding Academic Paper Award, Korea Institute of Building Construction Fall Conference 11/2025
Outstanding Presentation Paper Award, Architectural Institute of Korea Fall Conference 10/2024
Excellence Award, 9th National University Student CM Competition (Team Leader) 08/2024
Excellence Award, Safety Partnership Idea Competition (Team Leader) 05/2024
Participation Award, 8th National University Student CM Competition 08/2023
Outstanding Idea Award, Structural Seismic Design Competition 07/2023

7. LEADERSHIP AND COMMUNITY SERVICES

President of Architectural Engineering Student Association, Pusan National University ..... 03/2023 – 02/2024

8. COMPETENCES

Programming Language: Python
AI and Research Tools: PyTorch, YOLO, LLaVA, LoRA, Stable Diffusion/SDXL, image quality assessment, and LLM-based multi-agent workflows
Languages: Korean (native) and English (OPIc IH)
Other: MS Office